

Powerful & Generalizable, Why not both? VA: Various Attacks Framework for Robust Adversarial Training

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Abstract: Due to its effectiveness, adversarial training (AT) is becoming the first choice to improve the robustness of deep learning models against adversarial attacks. AT is formulated as a min-max optimization problem. The performance of AT is essentially reliant on the inner optimization problem (i.e., max optimization), which requires the generation of adversarial examples. Most AT methods rely on a single attack to craft these examples neglecting the impact of image-class robustness on the adversarial training. This oversight led to shortcomings such as poor generalization on both perturbed and clean data, unreliable robustness against unseen adversarial attacks, and limited exploration of the perturbation space. Therefore, an investigation and analysis of AT robustness via adapting various attacks based on *image-class robustness* is still unaddressed. In this paper, we propose Various Attacks (VA), a novel framework for a robust and generalizable adversarial training based on image-class robustness. Our framework introduces two novel components: Advanced Curriculum Training (ACT), which ensures the diversity of adversarial attacks by gradually increasing attack strength while rotating through these attacks, and Class-Attack Assignment (CAA), which adaptively determines and assigns the optimal adversarial attack to each image-class to maximize the loss. The proposed framework trains image classification neural networks using a variety of adversarial attacks that significantly improve the generalization robustness. The results of experiments on two benchmark datasets show the superiority of the VA framework over state-of-the-art adversarial training methods.

1 INTRODUCTION

Deep neural networks (DNNs) have achieved significant success in various domains such as image classification (Khamaiseh et al., 2022b), voice recognition (Kosuge et al., 2023), and security applications (Khamaiseh et al., 2022a) (Khamaiseh et al., 2020) (Alsmadi et al., 2016). However, it is been found that DNNs are vulnerable to adversarial examples (Szegedy et al., 2013). Adversarial examples can be generated by adding small, often imperceptible perturbations to clean images causing even well-trained image classification models to misclassify them. This vulnerability raises security concerns about adapting DNNs, particularly in critical applications. To address this issue, researchers have proposed various defense methods (Goodfellow

et al., 2015) (Madry et al., 2019) (Papernot and McDaniel, 2017) (Kurakin et al., 2016) (Liao et al., 2018). Among these, adversarial training (AT) stands out as the most effective method for enhancing the robustness of DNN models against adversarial attacks. Generally speaking, standard adversarial training involves incorporating adversarial examples during the training phase to improve the DNN's robustness. The AT framework is formulated as a min-max optimization problem, where generating effective adversarial examples (i.e., solving the inner maximization problem) is the key factor in determining the robustness of the AT methods.

Previous studies have revealed that discrepancies between image classes impact natural training (i.e., training on clean examples). Specifically, some image classes are harder to learn and require more time to converge compared to other classes (Tang et al., 2020) (Wang et al., 2019). In adversarial settings, a few people investigate the image-class robustness,

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and they reveal that some image classes show lower robustness than other classes against some adversarial attacks. Furthermore, examples within the same image class are often vulnerable to similar perturbations, causing them to be misclassified into the same incorrect class, as other correctly labeled images (Tian et al., 2021). However, in the adversarial training community, the majority of proposed AT defense methods (Cui et al., 2023)(Zhang et al., 2021)(Zhang et al., 2019a)(Addepalli et al., 2022)(Xu et al., 2023)(Huang et al., 2022) solve the inner maximization problem by using a single adversarial attack (e.g., PGD attack(Dong et al., 2018)). However, they often overlook the differences in image classes robustness and their impact on the generalization robustness of the adversarial training (Tang et al., 2020)(Wang et al., 2019). This oversight is partly due to the assumption that the benchmark datasets seem to have balanced robustness across all image classes. As detailed in §2, single-attack AT methods suffer from multiple shortcomings, with a significant issue being their poor generalization robustness against unseen adversarial attacks. Generalization robustness is a pivotal property that measures the ability of the AT defense method to provide robustness against unseen, newer, stronger, and adaptive adversarial attacks(Lee et al., 2020). The phenomenon of improving the AT generalization robustness by leveraging the discrepancies of image class robustness using multiple attacks is still largely unaddressed.

To fill this void, we propose the various attacks method (VA). VA is a novel framework that applies multiple adversarial attacks based on the robustness levels of different image classes leading to improving the overall model robustness. The VA framework comprises two components as follows: (1) the advanced curriculum training (ACT) systematically rotates a variety of adversarial attacks across image classes while gradually increasing the attacks’ strength, thereby improving the model’s generalization robustness. (2) The novel Class-Attack Assignment (CAA) algorithm exploits the discrepancies of image-class robustness to assign adversarial attacks to image classes at key training points in a way that maximizes the loss. To evaluate the performance of our proposed methods, we conduct extensive experiments using the CIFAR-10 and CIFAR-100 datasets. During our experiments, we tested state-of-the-art AT defense methods against several adversarial attacks. The reported results in §6 validate the effectiveness of our proposed framework in improving the overall robustness, including the generalization robustness of the DNN models when compared to other state-of-the-art AT defense methods.

To summarize, our contributions are as follows:

- We propose and implement a novel Various Attacks (VA)¹ framework that utilizes various adversarial attacks based on image class discrepancies to improve the overall robustness, including the generalization robustness, of trained DNN models against stronger and more adaptive adversarial attacks.
- We propose the Advanced Curriculum Training (ACT) method that can be viewed as an improved form of curriculum training in addition to the a new Class-Attack Assignment (CAA) algorithm. Jointly, these components provide an effective solution to the inner-maximization problem of AT. This method can be applied to other AT defense methods to improve the generalization robustness.
- We conduct extensive comparisons with state-of-the-art AT defense methods using CIFAR-10 and CIFAR-100 datasets and six other baseline adversarial attacks. The reported results demonstrate that the VA framework significantly improves the robustness of the trained DNN models and outperforms other AT defense methods.

The rest of the paper is organized as follows. In §2, we discuss the motivation of this work by introducing the shortcomings of the single-attack AT defense methods. In §3, we discuss the preliminary and the background of this work. An extensive overview of the related work is explained in §4. In §5, we present the formulation of the proposed framework. Experimental settings and results are explained in §6. Finally, §7 concludes our work.

2 MOTIVATION

Generalization Robustness of Single-Attack AT.

AT methods typically use a single attack throughout the training process to approximate the inner maximization problem. This step involves finding the adversarial example that causes the highest loss before adjusting the model weights through backpropagation to minimize its impact (outer minimization problem). However, relying on a single attack limits the exploration of the space of perturbations (Croce and Hein, 2020), resulting in a vulnerable trained model with poor generalization robustness (Lee et al., 2020). Even single-attack defense methods that utilize variable hyperparameters are optimized to defend only

¹The source code is available: <https://github.com/LAiSR-SK/VariousAttacks>.

against the specific types of attacks (e.g., gradient-based attacks) (Gowal et al., 2021). While such models can show high robustness against the attack used during training, they lack generalization robustness, making them vulnerable to other attacks. This problem is introduced in (Cai et al., 2018) and demonstrated in Table 1, which displays the robust accuracy of a model trained adversarially using the FGSM attack against multiple attacks. The FGSM-trained model achieves a high robust accuracy of 81.55% against the FGSM attack but fails against stronger attacks like PGD-20 (Madry et al., 2019) and AutoAttack (AA) (Croce and Hein, 2020), with robust accuracies of only 0.04% and 0.0%, respectively. To address these shortcomings, recent defense methods have adapted the PGD attack with random initialization to increase the diversity of perturbations and improve the model’s generalization robustness and overall robust accuracy (Croce and Hein, 2020) (Croce and Hein, 2021). However, such defense methods have still been defeated by stronger and more adaptive attacks (Dong et al., 2020). Another example of a single-attack training method is Customized-AT (Cheng et al., 2020), also shown in Table 1. While it shows a high accuracy against most attacks, it is defeated by AA with a very low robust accuracy of 21.68%, indicating poor generalizability to more sophisticated and adaptive attacks. These examples demonstrate that models trained with a single attack lack the defensive capabilities necessary for them to be considered robust against the wide variety of adversarial attacks, underscoring the need for more diverse and robust training methods.

Table 1: The robust accuracy values of two WideResNet-34 models on the CIFAR-10 dataset. One model is trained using the FGSM attack and another one is trained using the Customized-AT defense method. The table compares their performance against multiple attacks.

Single-Attack Training Accuracy		
Attack	FGSM	Customized-AT
Clean	85.59%	94.04%
PGD-20	0.04%	68.47%
MIM	0.0%	74.13%
FGSM	81.55%	81.29%
AA	0.0%	21.68%

3 PRELIMINARIES

3.1 Adversarial Training (AT)

As depicted in Equation 1, AT is formulated as a min-max optimization problem:

$$\min_{\Theta} \mathbb{E}_{(x,y) \in D} [\max L(f_{\Theta}(x + \delta), y)] \quad (1)$$

where f_{Θ} represents the model with parameters Θ , x is an input image and y its assigned image-class in dataset D , and $L(\cdot)$ is a loss function that measures the error between the model’s prediction and the true class. $\mathbb{E}_{(x,y) \in D}$ denotes the expected value of the inner maximization problem for a sample (x, y) from D . The goal of AT is to find the parameter set Θ of f that minimizes the expected maximum loss $\mathcal{L}$ (for an image (x, y) in dataset D) caused by a perturbation δ . δ is constrained to ensure the imperceptibility of adversarial images to the human eye and prevent the true class of the targeted image from shifting. In other words, AT optimization problem is often divided into two subproblems: the *outer minimization* and *inner maximization* problems. While the outer minimization problem is traditionally solved using gradient descent to find the optimal model parameters, a large portion of recent research is focused on how to approximate the solution for the inner maximization problem (Madry et al., 2019).

3.2 Class-Wise Properties

Some works (Tang et al., 2020)(Wang et al., 2019) reveal that certain classes are inherently harder to learn than other classes, usually refer these type of class as *hard-classes*. In adversarial training, these classes have a lower image-class robustness (i.e., vulnerable classes), making them more susceptible to attacks, even after AT training. The problem is even more prevalent in adversarial training compared to standard training, as demonstrated by the *class-wise variance* (Tian et al., 2021). Letting $|C|$ represent the number of classes in a dataset, α_y denote the accuracy of class y , and $\bar{\alpha}$ denote the average accuracy across all classes, the CV is defined as:

$$CV = \frac{1}{|C|} \sum_{y=1}^{|C|} (\alpha_y - \bar{\alpha})^2 \quad (2)$$

In (Tian et al., 2021), authors reported that class-wise variance of adversarial training is 28 times higher than in standard training. This increased variance causes vulnerable image-classes to reduce the efficacy of adversarial training remarkably. Therefore, it becomes essential to enhance the image-class robustness of the most vulnerable classes to improve the overall robustness of the model. This can be done by targeting specific vulnerable classes and improving the robustness at the weakest points of the dataset. This approach involves maximizing the loss generated through perturbing images within these vulnerable image classes. This targeted approach not only increases the robustness of the most vulnerable image-classes, but also contributes to the overall robustness of the model.

Another significant property of class-wise is *class grouping*. Recent research has shown that image-classes can often be divided into groups called superclasses (Tian et al., 2021). An image from an image-class within one superclass is more commonly perturbed to be classified as another image-class within that superclass, and less commonly perturbed to be classified outside its group. For instance, a “car” image within the “vehicle” superclass is more likely to be misclassified as a “truck” within the “vehicle” group, rather than as a “dog”, which is outside the group. Images that fall under the same class or superclass possess similar characteristics that can be leveraged in the AT process to enhance the accuracy of the model (Tian et al., 2021).

4 RELATED WORK

Adversarial Attacks. Many adversarial attack methods have been proposed to exploit vulnerabilities in neural networks (Khamaiseh et al., 2022b). The Fast Gradient Sign Method (FGSM) attack (Goodfellow et al., 2015) calculates the loss gradient for each image and perturbs the target image in that direction. Building on FGSM, the Projected Gradient Descent (PGD) attack iterates FGSM for a set number of steps. After each perturbation step the attack projects the adversarial image x_{adv} back onto the ϵ -ball around the clean image, ensuring the perturbation remains small and imperceptible. Dong et al. (Dong et al., 2018) enhance this approach with the Momentum Iterative Method (MIM) attack, which adds a momentum term and uses previous gradients to update x_{adv} at each iteration. Carlini and Wagner (CW) (Carlini and Wagner, 2017) propose an attack that creates adversarial images by adjusting the standard objective function, specifically designed to evaluate defense methods. The Target-X attack (Khamaiseh et al., 2023) gradually builds up the least perturbations by calculating the projection onto a specified decision boundary and adding a tuning vector to force a misclassification. Finally, Croce et al. (Croce and Hein, 2020) propose AutoAttack (AA), a sequence of up to six different attacks designed to test models’ robustness. AA is widely regarded throughout the research community as a benchmark for evaluating the robustness of adversarial training (AT) methods.

Adversarial Training. The AT optimization problem (see Equation 1) is an NP-hard problem. This is due to the inner maximization problem: the cost function of neural networks are non-concave (and non-convex) with respect to the input (Huang et al., 2022) (Madry et al., 2019). The standard AT approach, proposed in

(Madry et al., 2019), uses a PGD attack to approximate the inner maximization problem. This approach has been extended and adjusted in various ways, such as adjusting the loss function, altering the framework structure, and adapting strategies to evolve with specific images or training stages.

Novel Loss Functions. Standard AT training typically uses a predefined loss function such as Cross-Entropy (CE) or Kullback-Leibler (KL) divergence to calculate the distance between the model output and the correct labels. Cui et al. (Cui et al., 2023) propose an Improved Kullback-Leibler (ILK) loss, which addresses asymmetry by enabling gradient propagation and replaces sample-wise weights with class-wise weights. Pang et al. (Pang et al., 2022) introduced the SCORE robust error which calculates the loss as the maximum distance between the data distribution and the model output for adversarial images. Zhang et al. (Zhang et al., 2019b) proposed the TRADES loss function, which adds an additional hyperparameter to control the balance between clean accuracy and robustness. These loss functions are commonly used in conjunction with other methods to boost robustness (Jia et al., 2022) (Wu et al., 2020) (Zhang et al., 2021). However, they fail to consider the class properties we exploit in the Various Attacks (VA) method. Moreover, these loss functions remain unchanged throughout the AT process, which means they neglect the weak areas of training and fail to adapt to specific images or stages of training (Tian et al., 2021). Simply adjusting the loss function while keeping other factors constant throughout reduces the generalization robustness of the final model. This is because the training cannot adaptively respond to the changing capabilities of the model and the properties of the dataset.

Adaptive Training Strategies. These AT formulations change the training strategy throughout the process for a more effective solution. Zhang et al. (Zhang et al., 2021) propose GAIRAT, which categorizes samples as “attackable” or “guarded” based on their vulnerability to attack, adjusting weights accordingly. Xu et al. (Xu et al., 2023) propose DyART, which prioritizes the increase of distances between samples and decision boundaries using the novel Temperature-PGD attack. Tian et al. (Tian et al., 2021) leverage class-specific properties of different examples to increase a model’s robustness. Cai et al. (Cai et al., 2018) propose Curriculum AT (CAT), increasing attack strength as training progresses. Shi et al. (Shi and Liu, 2024) further analyze curriculum training and provide a mathematical foundation for the success of adaptive training methods. Jia et al. (Jia et al., 2022) design the LAS-AT framework, utiliz-

ing a secondary model to generate a targeted attack strategy at each training stage. These methods aim to increase robustness against a wider variety of adversaries. However, they lack the variety of attacks that the VA method employs to improve generalization robustness. Additionally, they adapt training based only on either the stage of training or image-specific properties, whereas the VA framework adjusts training strategies based on both aspects. As a result, these methods produce models with lower robust accuracy values and less generalization robustness than VA, as we demonstrate in §6.

5 VARIOUS ATTACKS FRAMEWORK (VA)

Here, we introduce the VA (Various Attacks) adversarial training method, designed to increase both the overall model robustness and generalizability robustness. The VA framework takes advantage of the class-wide differences discussed in §3.2 by assigning specific attacks to specific classes based on the relative attack strength. In this context, the term *assignment* means that for the image-class y and attack a at epoch e , attack a is used to perturb the images in class y during that specific training epoch. These assignments are epoch-specific and are adjusted throughout training to improve the generalization robustness. They maintain the diversity of generated examples while still maximizing loss values to effectively approximate the solution to the inner maximization problem in AT.

The VA framework consists of two major components. First, Advanced Curriculum Training (ACT) gradually incorporates stronger attacks over the course of training while maintaining the ability to pinpoint the most vulnerable classes at each stage. This approach helps in progressively strengthening the model against increasingly powerful adversarial examples. Second, the Class-Attack Assignment (CAA) algorithm assigns the strongest attacks to the most vulnerable classes at key training stages, thereby increasing the lowest image-class robustness (i.e., improving the robustness of the weakest image-classes) and consequently improving the overall robustness of the model.

5.1 Advanced Curriculum Training

Curriculum training is an AT technique that gradually increases the strength of the attacks used to generate adversarial examples. The rationale behind this approach is that stronger attacks will yield more ef-

fective solutions to the inner maximization problem of AT, thereby theoretically leading to a more effective training process and improving model robustness (Madry et al., 2019). However, previous work has highlighted a significant drawback: models trained exclusively with stronger attacks tend to “forget” how to defend against weaker attacks. This phenomenon leads to decreased accuracy against such less effective attacks, thereby reducing overall model robustness (Cai et al., 2018). Curriculum training works by adaptively adjusting the training radius to correspond with the Wasserstein distance between the adversarial distributions of adjacent training iterations. In the earlier phases of training, a small radius (indicating a weaker attack) is used to stabilize the model when the distance between iterations is relatively large. In later phases and as training progresses, the radius is gradually increased to incorporate stronger attacks and improve robustness as the distance decreases (Shi and Liu, 2024).

Despite its success in improving the robustness of adversarially trained models against certain attacks, curriculum training still suffers from several problems similar to traditional AT. First, it relies on a single attack type adjusted to various strength levels to generate adversarial examples throughout the training process. Although this approach helps a model to develop increased robustness against different strength levels of the attack used during training, it does not help enhancing the model’s generalization robustness against attacks not used in training. Moreover, the existing form of curriculum training does nothing to specifically target vulnerable image-classes during training, the presence of which have been proven to reduce the overall robustness of a model. To address these issues we propose Advanced Curriculum Training (ACT), an improved form of curriculum training framework designed to enhance generalization robustness by incorporating multiple attack types throughout the training phases. ACT employs various attacks at each training phase while gradually increasing attack strength throughout the training process. This method maintains the benefits of standard curriculum training while diversifying the types of attacks used, thereby improving the model’s generalization robustness. ACT is composed of two key components: phase-based curriculum training and class revolution.

Phase-Based Curriculum Training. Phase-based curriculum training divides training into multiple *phases*, or groups of subsequent epochs. During each phase E , a set of attacks $\mathcal{A}$ with similar strengths is assigned to generate adversarial examples for different image classes. To determine the strength of an attack,

we examine the loss values produced by each attack when perturbing a set of clean images. Higher average loss values indicate stronger attacks and lower loss values indicate weaker attacks. Using a variety of attack types across successive training iterations prevents the model from overfitting to a single attack type. Instead, it forces the model to adapt to various distributions of adversarial examples. This approach maintains the benefits of standard curriculum training, with the relative strength of attacks increases throughout the training and adjusts according to the Wasserstein distances between distributions of consecutive training iterations. Simultaneously, it raises the generalization robustness by incorporating a broader diversity of attacks during training.

Class Revolution. To maintain the attack diversity, ACT employs a mechanism called *class revolution* for assigning attacks to classes during phase-based curriculum training. Within each training phase E , every attack $a \in \mathcal{A}$ is systematically rotated through all classes. This ensures that each attack in $\mathcal{A}$ is used to generate adversarial examples for every image class in the dataset for at least one epoch. The strongest attack-class pairings, determined using the CAA algorithm are reserved for the final assignment in each phase.

Class revolution ensures that the generalization robustness of individual classes remains high. As discussed in §3.2, image-classes within datasets have different properties and thus different levels of robustness. Equation 3 defines the *class-wise variance of generalization robustness*:

$$CV_{GR} = \frac{1}{C} \sum_{c=1}^C \frac{1}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} (\alpha_{c-a} - \bar{\alpha}_a)^2 \quad (3)$$

where α_{c-a} represents the accuracy of class c under attack a , and α denotes the average accuracy across all classes for attack a . A high class-wise variance of generalization robustness indicates that the more vulnerable image classes are significantly reducing the overall image-class robustness. This occurs when each image class is only exposed to certain attacks, resulting in a lower generalization robustness. Class revolution mitigates this issue by ensuring that each image class is subjected to all available attacks, thereby enhancing the generalization robustness of each class. A small-scale variation of curriculum training is implemented within each phase as the strength of class-attack assignments increases until the strongest pairings have been applied. This comprehensive exposure reduces the class-wise variance of generalization robustness while increasing the overall generalization robustness of the model.

5.2 Class-Attack Assignment Algorithm

The Class-Attack Assignment (CAA) algorithm determines the strongest class-attack assignments used in the final pairing of each training phase. Using loss values calculated prior to training, the CAA iteratively assigns the strongest attacks to generate adversarial examples from images in the most vulnerable image-classes. Once an attack is assigned, it is removed from consideration to ensure diversity in adversarial examples. Analogous to how the strongest attacks are set, the most vulnerable image classes are identified by looking at the average loss values caused by all attacks. The class with the highest average loss value is the most vulnerable, while image-classes with lower average loss values are less vulnerable. By placing the strongest assignments last in each phase, the CAA maximizes its impact on increasing the robustness values of the most vulnerable image classes. In the final training phase, the model is trained on the strongest class-attack assignments for the strongest attacks.

Working Process of CAA. As depicted in Algorithm 1, the CAA starts calculating the average loss $\bar{l}_y$ using all selected adversarial attacks for each image class y (lines 1-3) via the formula below:

$$\bar{l}_y = \frac{1}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} \frac{1}{|\mathcal{L}_{y-a}|} \sum_{i=1}^{|\mathcal{L}_{y-a}|} l_{y-a_i} \quad (4)$$

where $\mathcal{L}_{y-a}$ denotes the set of loss values calculated by using attack a against class y , and l_{y-a_i} represents a specific loss value within the set. On (line-4), the image-classes are sorted in descending order based on $\bar{l}_y$. On (lines 7-14), the CAA algorithm iterates through each class, starting with the class y with the highest $\bar{l}_y$. it calculates the average loss $\bar{l}_{y-a}$ of each attack a against class y as follows:

$$\bar{l}_{y-a} = \frac{1}{|\mathcal{L}_{y-a}|} \sum_{i=1}^{|\mathcal{L}_{y-a}|} l_{y-a_i} \quad (5)$$

The attack with the highest $\bar{l}_{y-a}$ is selected and assigned to y (line -11):

$$a_y \leftarrow \operatorname{argmax}_a \bar{l}_{y-a} \quad (6)$$

The selected attack a is then removed from consideration for other classes (line- 12). This process repeats for the class with the next highest $\bar{l}_y$, so that it is assigned to the remaining attack with the highest $\bar{l}_{y-a}$. This continues until all classes have been assigned a different attack.

5.3 Theoretical Formulation

In this section, we formally define the loss and objective functions of the VA method.

Algorithm 1: Class Assignment Algorithm.

Input : L – Loss values for each attack for every class
 C – Classes
 $\mathcal{A}$ – Attacks

Output: $\mathcal{A}_{final}$ – Unique attack assigned to each class

```

1 for class  $y \in C$  do
2    $\bar{l}_y = \frac{1}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} \frac{1}{|L_{y-a}|} \sum_{i=1}^{|L_{y-a}|} l_{y-a_i}$ 
3 end
4  $C_{sorted} \leftarrow C$ 
5  $\mathcal{A}_r \leftarrow \mathcal{A}$  // Remaining attacks
6  $\mathcal{A}_{final}$ 
7 for class  $y \in C_{sorted}$  do
8   for attack  $a \in \mathcal{A}_r$  do
9      $\bar{l}_{y-a} = \frac{1}{|L_{y-a}|} \sum_{i=1}^{|L_{y-a}|} l_{y-a_i}$ 
      // Calculate average loss for each a left for c
10  end
11   $a_y \leftarrow \operatorname{argmax}_a \bar{l}_{y-a}$ 
      // Select remaining attack with highest loss
12   $\mathcal{A}_r \leftarrow \mathcal{A}_r - a_y$ 
13   $\mathcal{A}_{final} = \mathcal{A}_{final} + a_y$ 
14 end
15 Return:  $\mathcal{A}_{final}$ 

```

Sample-Specific Loss Definition. Each attack $a \in \mathcal{A}$ is provided with a separate bound to ensure perturbations remain appropriately small. If perturbations are unbounded, AT cannot fit all samples and is forced to sacrifice some, leading to a distorted decision boundary and diminished robustness (Cheng et al., 2020). We use ε_a to denote the perturbation limit for attack a and then define the loss for sample $(x, y) \in D$ as the following:

$$\max_{a_y(x) \in B(x, \varepsilon_{a_y})} \mathcal{L}(f_{\Theta}(a_y(x)), y) \quad (7)$$

Here, a_y denotes the attack assigned to image-class y at the current training stage. $B(x, \varepsilon_a)$ represents the l_p -norm ball centered at x with radius ε_a . This formula selects the perturbed image $a_y(x)$ within the ε_a -ball around x with the largest loss.

Class-Wise Loss Definition. As described earlier, an example x is assigned an attack a_y based on its correct image-class y . As such, we define the class-wise loss for an image-class y as:

$$\sum_{a \in \mathcal{A}} \sum_{x \in D_y} \mathcal{L}(f_{\Theta}(a_y(x)), y) \quad (8)$$

Here, D_y describes the subset of dataset D with correct the image-class y . This formula calculates the loss for all images x with image-class y when perturbed by every attack a from the attack set $\mathcal{A}$. (Recall that by using class revolution in ACT, the VA method

trains samples against each available attack at some stage of training).

Formal Formulation. Based on the sample-specific and class-wise loss definitions defined above, along with the standard AT formulation defined in Equation 1, the VA objective function is defined as:

$$\min_{\Theta} \mathbb{E}_{(x,y) \in D} \frac{1}{|C|} \sum_{y \in C} \left[\frac{1}{|D_y|} \sum_{a \in \mathcal{A}} \sum_{x \in D_y} \max_{a_y(x) \in B(x, \varepsilon_{a_y})} \mathcal{L}(f_{\Theta}(a_y(x)), y) \right] \quad (9)$$

where C denotes the number of classes in dataset D .

5.4 VA Pipeline

The VA framework is depicted in Algorithm 2. The attacks assigned to each class at a particular epoch are determined prior to training using the ACT and CAA techniques described above. For each batch, the average image-class is identified and the attack is selected accordingly (lines 4-5). Then, the images in the batch are perturbed using the selected attack (line 6). The loss between the correct image-class and the model output of the perturbed image is calculated (line-7). Finally, the calculated loss is used to update the model parameters (line 8).

Algorithm 2: The Pipeline of Various Attacks.

Input : N – Number of epochs
 f_{Θ} – model with parameters Θ
 η – Learning rate
 $\mathcal{A}_{final}$ – Assignments
 D – Dataset

```

1 Initialize  $\Theta$ 
2 for epoch = 1 to  $N$  do
3   for batch  $b \subset D$  do
4      $\bar{y} \leftarrow \operatorname{argmax} f_{\Theta}(b)$ 
5      $a \leftarrow \mathcal{A}_{final-\bar{y}}$ 
      // Select attack based on class
6      $x_{adv} \leftarrow a(x)$  // Perturb x with attack a
7      $l \leftarrow \mathcal{L}(f_{\Theta}(x_{adv}), y)$ 
8      $\Theta \leftarrow \Theta - \eta \cdot \nabla l$ 
9   end
10 end

```

5.5 Attack Selection Criteria

For the VA algorithm to be most effective, it is important to select attacks that provide diversity and meet criteria related to practical feasibility in training.

Attack Diversity. To ensure diversity, the chosen set of attacks must incorporate distinct elements and strategies for perturbing an image. This may include

using different norms for distance calculations, different loss functions, randomness, and/or adding additional terms to gradient calculations. The focus here is on having a set of differences rather than on the nature of the differences themselves. Proper attack diversity results in a diverse set of generated adversarial examples (Croce and Hein, 2020). Therefore, the proper selection of attacks is essential to the success of the VA method — using attacks that use similar perturbation methods will not generate the necessary diversity in adversarial examples, which is necessary for effective training.

Practical Feasibility. The chosen attacks must be suitable for practical use in adversarial training settings. This can be determined by three criteria: time, space, and effectiveness. Each attack should be fast and require minimal computational resources, as slow attacks that demand extensive resources such as disk and GPU space make the training process impractically long or unfeasible. Additionally, the selected attacks must be effective in increasing generalization robustness during AT. Ineffective or weak attacks will negatively impact the model’s performance. As detailed in §6.5, we only select attacks that are experimentally proven to improve model robustness when used in AT.

6 EXPERIMENTS

This section details our experimental setup, including the adversarial attacks and defense methods used. Then, we provide an extensive analysis of the results of VA compared to other AT methods. Finally, we provide an ablation study of the VA framework.

6.1 Experimental Design

Comparison with Existing Methods. To provide a comprehensive evaluation, we compared our VA framework with the baseline as well as a variety of reputable and highly cited AT defense methods. We utilized the source code and the recommended training settings provided by the authors to train models using the following methods: GAIRAT, ADT, TRADES, LAS-AT, and FAT (Friendly Adversarial Training) (Zhang et al., 2020). In addition, we trained models adversarially using the PGD attack as proposed in (Madry et al., 2019) (labeled as ‘Standard AT’) and trained models only on clean data with no adversarial training or other defenses applied (labeled as ‘Clean’). We also tested DNR (Kundu et al., 2021) and YOPO (Zhang et al., 2019a) using the pre-trained models provided by the authors.

Evaluation Metrics. We evaluated all models against the AA attack using the standard evaluation set (APGD-T, APGD-DLR, FAB, and Square Attack) with ℓ_∞ norm and $\epsilon = 8/255$. Also, we tested the models against the ℓ_∞ norm PGD attack with 20 and 40 steps, a step size of 0.01, and $\epsilon = 8/255$. Additionally, we evaluated the models against the CW, MIM, and FGSM attacks using 20 steps with $\epsilon = 8/255$, a step size of $2/255$, and a margin of 50 where applicable. Then, we reported two metrics: *clean accuracy* and *robust accuracy*. Clean accuracy refers to the model accuracy when tested against unperturbed samples from the predefined test subset of the dataset. Robust accuracy refers to the accuracy of the model when tested against samples from the same test subset but perturbed with the attack under evaluation.

6.2 Settings

We implemented all experiments using Pytorch 2.01 with CUDA enabled and conducted on NVIDIA GeForce RTX 4090 GPUs. We used a batch size of 128 for the VA method and trained for 100 epochs on CIFAR-10 and 110 epochs on CIFAR-100. The training included a warm-up round of 15 epochs for CIFAR-10 and 25 epochs for CIFAR-100. We set the weight decay to $2e-4$ and used an initial learning rate of 0.1 with a 10% decay at the 75th, 90th, and 100th epochs. All training is conducted using an SGD optimizer with a momentum of 0.9. We trained VA models using the WideResNet-34 architecture.

6.3 Robustness Comparisons

CIFAR-10 Dataset. As shown in Table 2, the VA framework achieved higher robust accuracy scores than the majority of other AT methods. Specifically, VA outperformed Standard-AT, TRADES, ADT, GAIRAT, LAS-AT, DNR (C), DNR (I), YOPO, and FAT in terms of robust accuracy across all attacks. (The results of the Curriculum AT and Customized AT methods are discussed in §6.4). The VA method also achieved 91.15% clean accuracy, which is higher than all methods except Customized AT. The clean model without any AT training achieves 95.09%. This discrepancy can be attributed to the accuracy-robustness trade-off, where AT generally decreases clean accuracy (Lee et al., 2020).

CIFAR-100 Dataset. The results on the CIFAR-100 dataset depicted in Table 3, further support the superiority of the VA method. It achieved a higher robust accuracy against all attacks compared to Standard AT, TRADES, ADT, GAIRAT, LAS-AT, YOPO, and FAT. (The results of the Curriculum AT and Customized AT methods are discussed in §6.4).

Table 2: The robustness of AT defense methods on the CIFAR-10 dataset. Bold results with (*) refer to the highest results among all AT methods. Bold results without (*) refer to the highest results among the majority of other AT methods.

The Experimental Results of CIFAR-10 Dataset							
Defense Method	Clean	FGSM	MIM	CW	PGD-20	PGD-40	AA
Clean (Without AT)	95.09%*	26.08%	0.00%	0.00%	0.00%	0.00%	0.00%
Standard AT	86.89%	57.78%	51.09%	49.90%	51.68%	51.56%	48.32%
TRADES	84.58%	60.18%	54.97%	52.95%	55.53%	55.40%	52.02%
ADT	83.63%	56.90%	49.93%	48.73%	50.51%	50.29%	45.98%
GAIRAT	85.74%	56.69%	56.81%	44.48%	58.63%	58.67%	42.48%
LAS-AT	87.34%	62.11%	55.81%	54.72%	56.39%	56.23%	53.03%
DNR (C)	87.48%	55.74%	46.65%	44.76%	47.41%	47.00%	42.40%
DNR (I)	87.31%	54.69%	45.80%	43.07%	46.42%	46.18%	40.97%
YOPO	86.34%	55.26%	48.17%	47.71%	48.72%	48.37%	44.93%
FAT	89.06%	58.81%	48.78%	47.29%	48.28%	47.96%	44.42%
Curriculum AT	89.92%	78.55%*	3.83%	35.40%	40.27%	26.01%	0.14%
Customized AT	94.04%	81.29%*	74.13%*	58.79%*	68.47%*	66.40%*	21.68%
Ours (VA)	91.15%	64.98%	61.69%	56.52%	68.71%*	68.58%*	55.74%*

Table 3: The robustness of AT defense methods on the CIFAR-100 dataset. Bold results with (*) refer to the highest results among all AT methods. Bold results without (*) refer to the highest results among the majority of other AT methods.

The Experimental Results of CIFAR-100 Dataset							
Defense Method	Clean	FGSM	MIM	CW	PGD-20	PGD-40	AA
Clean (Without AT)	78.65%*	8.77%	0.01%	0.00%	0.00%	0.00%	0.00%
Standard AT	60.73%	31.08%	27.47%	26.13%	27.79%	27.64%	24.90%
TRADES	52.06%	27.88%	25.79%	22.75%	26.52%	26.53%	21.93%
ADT	57.72%	30.50%	24.76%	23.88%	25.47%	25.29%	21.53%
GAIRAT	60.06%	28.61%	24.66%	23.11%	25.08%	25.01%	21.28%
LAS-AT	59.22%	32.00%	26.39%	23.21%	25.75%	25.45%	21.96%
YOPO	62.31%	28.51%	24.23%	23.57%	24.48%	24.31%	21.37%
FAT	65.09%	29.18%	23.24%	23.02%	23.25%	23.14%	21.44%
Curriculum AT	64.73%	70.55%*	0.98%	8.92%	20.32%	12.97%	0.03%
Customized AT	73.14%	45.99%*	37.23%*	7.23%	34.96%*	33.83%*	11.59%
Ours (VA)	61.90%	32.77%	29.55%	28.13%*	29.92%	30.11%	25.93%*

tomized AT methods are discussed in §6.4). Additionally, the clean accuracy for VA on CIFAR-100 is equal to 61.90%, which is higher than many AT methods including Standard AT, LAS-AT, ADT, GAIRAT, and TRADES.

6.4 Analysis of Generalization Robustness

The results reported in Tables 2 and 3 demonstrate the VA method’s ability to generalize to a variety of attacks, thereby increasing the robustness of the trained model against attacks of different types and strengths. VA provides effective robustness against weak and strong types of gradient-based attacks such as FGSM and PGD. It is also able to defend a trained model against sophisticated and more adaptive attacks such as AA. Meanwhile, other AT methods show promising results in terms of clean and robust accuracy against some types of attacks. However, they fail to defend the trained models against more adaptive and stronger attacks. For instance, the Customized-AT defense achieved a higher robust accuracy on CIFAR-10 than the VA method when tested against FGSM, MIM, CW, and PGD-7 at-

tacks. It also achieved a higher clean accuracy. However, when the strength of the attack increased (e.g., PGD-20 and PGD-40), Customized-AT failed significantly and achieved lower robust accuracy than the VA method. Notably, Customized-AT only achieved 21.68% robust accuracy against AA, significantly lower than VA (55.74%) and the majority of other AT methods. The same pattern holds for CIFAR-100, where Customized-AT achieved high robust accuracy against weak attacks such as FGSM (45.99%). Meanwhile, it reported very low robust accuracy against AA (11.59%). This indicates that Customized-AT is designed to provide robustness against weak gradient-based attacks and cannot generalize to unseen and stronger attacks.

This lack of generalization robustness is also present in Curriculum-AT. On CIFAR-10, it reported a higher robust accuracy against the FGSM attack (78.55%) compared to the VA accuracy (64.98%). However, it achieved lower robust accuracy against the remaining attacks. Specifically, it reported only 0.14% robust accuracy against AA. This indicates a major weakness in the training process that leaves the model vulnerable to unseen attacks — a low generalization robustness. The results were corroborated on CIFAR-100 where the Curriculum-AT method re-

ports a higher robust accuracy against the FGSM attack, but a significantly lower robust accuracy against all other attacks. For instance, against both MIM and AA, Curriculum-AT reported a robust accuracy of less than 1%. Meanwhile, VA achieved 29.55% against MIM and 25.93% against AA.

These results demonstrate examples where certain AT defense methods fail to generalize to different types of attacks. Unlike these methods, the VA method significantly enhances overall robustness and specifically generalization robustness. By achieving the highest robust accuracy against the strong and adaptive AA attack on both CIFAR-10 and CIFAR-100 datasets, VA not only demonstrates its superior capability in defending DNN models against unseen, strong, and adaptive attacks but also stands out as a robust baseline AT method for ensuring model protection.

6.5 Ablation Studies

The parameters discussed in Section 6.2 were chosen using a combination of experimentation and previously established baselines. The weight decay, initial learning rate, choice of optimizer, and momentum were selected as the same training settings as the majority of the other tested methods in order to provide a more accurate baseline comparison. In this section, we describe the experiments we performed to select the attacks used in training, the appropriate length of training, and the most effective learning rate schedule.

The Impact of Attack Selection. Numerous techniques have been designed to attack DNN models, but many of them are not suitable for use in AT. To identify the most effective attacks for the VA framework, we conducted ablation studies by experimenting with different attacks and analyzed their impact on model robustness. For each experiment, we trained two models using the VA framework. All attacks and other training parameters are kept identical, except one model is trained using the attack under examination and the other is trained without it. Figure 1 demonstrates a sample of attacks and their impact on the AT robustness. This experiment concluded that the CW and MIM attacks increase both clean and robust accuracy, making them suitable for training. Conversely, the FGSM attack was found to significantly decrease both clean and robust accuracy, indicating its unsuitability for AT.

The Impact of Training Duration. We explore the impact of the number of training epochs on final model accuracy and robustness. While other AT methods can utilize *early stopping*, where training concludes after a set accuracy has been reached or

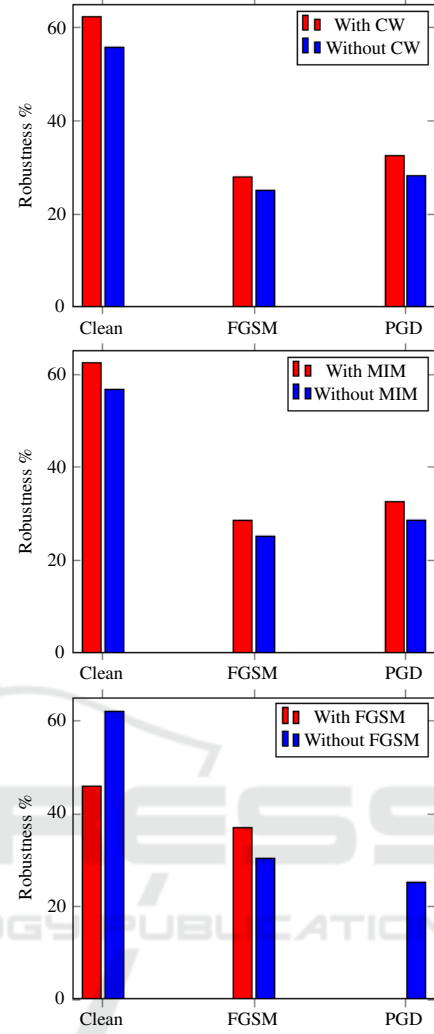


Figure 1: Three ablation studies, each comparing two models trained with or without a specific attack on the CIFAR-100 dataset.

progress has slowed to a set rate, the VA method requires all phases to be completed for maximum training success. Fig. 2 shows the best results from three different models trained on the CIFAR-100 dataset. One model was trained for 70 total epochs, another for 110, and the third for 140 epochs. The total number of epochs is divided into four phases of approximately 17-18, 27-28, and 35 epochs each. The attack choices, schedules, and other experimental settings are otherwise identical.

For all three categories, clean and robust accuracy increases when moving from 70 to 110 total epochs, indicating that a longer length of training could be useful. However, the clean and robust accuracy dropped when the model was trained for 140 epochs, reporting scores lower than the other two models in all three categories. This indicates that

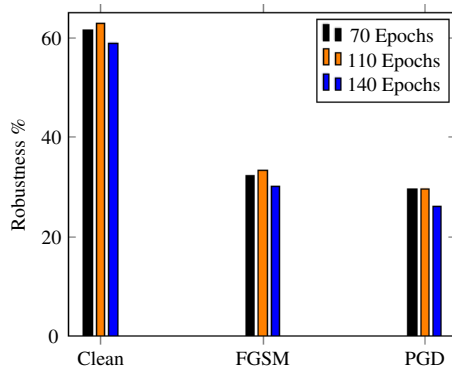


Figure 2: Ablation study comparing three WRN-34 models trained for differing lengths of training on the CIFAR-100 dataset.

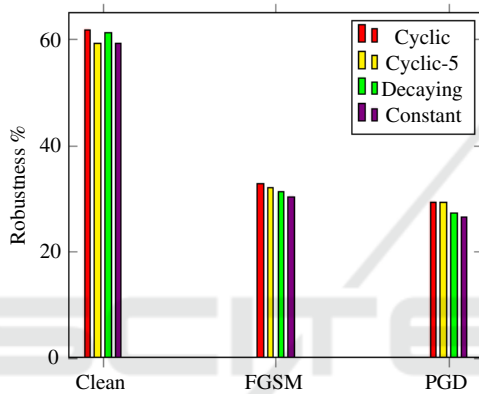


Figure 3: Ablation study comparing three WRN-34 models trained with different learning rate schedules on the CIFAR-100 dataset.

longer training does not always correlate with improved AT results. For each dataset, we selected the training length that resulted in the highest accuracies and robustness: 110 epochs for CIFAR-100 and 100 epochs for CIFAR-10.

The Impact of Learning Rate Schedule on AT. To select an effective schedule for adjusting the learning rate, we conducted ablation studies examining the effects of different learning rate schedules. Fig. 3 displays the results of four different learning rates. The ‘Cyclic’ model uses a learning rate that increases and decreases within each group of subsequent epochs. The ‘Cyclic-5’ model uses a learning rate that cycles every 5 epochs. The ‘Decaying’ model uses a learning rate that decays towards the end of training. The ‘Constant’ model uses a constant learning rate throughout training. All four models used a base learning rate of 0.1 and all other training settings were identical. This experiment demonstrated that the ‘Cyclic’ model achieved the highest clean accuracy and the highest robust accuracy against both PGD and

FGSM. Thus, we adopted the ‘Cyclic’ learning rate learning rate for the final training.

7 CONCLUSION

We propose the various attacks Various Attacks (VA) method, a novel adversarial training (AT) framework designed to enhance both overall and generalization robustness of models against unseen attacks. VA uses Advanced Curriculum Training (ACT) to adjust training at each stage dynamically, coupled with the Class-Attack Assignment CAA) algorithm to maximize loss effectively. By employing a variety of attacks, we the VA method significantly improves the generalization robustness of the models and overall performance. We further provide a theoretical formulation and general algorithm for the VA method as well as guidelines for the effective choice of attacks.

Our extensive experiments on two benchmark datasets demonstrate the effectiveness and generalizability of the VA method. The results demonstrate that VA significantly improves both the overall robustness and generalization robustness. Notably, it achieves the highest robust accuracy against AutoAttack (AA), an evaluation method specifically designed to test robustness against diverse attack types, thereby underscoring the superior generalization robustness of the VA method. The success of our method sheds lights on the limitations of current AT techniques, which often fail to protect models against unseen and stronger attacks, and position VA as a robust baseline for defending deep neural networks.

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